Phase 1 - Evidence - References used to inform the V0.0 material

Version 0.0

Introductory note

This is a list of books, articles, and other sources that inspired version 0.0 of the tasks and responsibilities of theorists in psychology. It is a cumulative body of literature rather than a set of references assigned to individual tasks: the eight responsibilities in version 0.0 were formulated by drawing on this literature as a whole.

Version 0.0 of the tasks and responsibilities was revised after the consensus review. The current version, version 0.1, also reflects the feedback of the ECTO experts and the sources and perspectives they brought in. This list therefore documents where the responsibilities started, not everything that informs their present wording.

References

Borsboom, D. (2013, November 20). *Theoretical amnesia.* Center for Open Science. http://osc.centerforopenscience.org/2013/11/20/theoretical-amnesia

Borsboom, D., van der Maas, H. L. J., Dalege, J., Kievit, R. A., & Haig, B. D. (2021). Theory construction methodology: A practical framework for building theories in psychology. *Perspectives on Psychological Science, 16*(4), 756–766. https://doi.org/10.1177/1745691620969647

Birnbaum, M. H. (1999). Testing critical properties of decision making on the Internet. Psychological Science, 10(5), 399-407. https://doi.org/10.1111/1467-9280.00176

Bringmann, L. F., Elmer, T., & Eronen, M. I. (2022). Back to basics: The importance of conceptual clarification in psychological science. *Current Directions in Psychological Science*, *31*(4), 340-346. https://doi.org/10.1177/09637214221096485

Chang, H. (2004). *Inventing temperature* (pp. 182-97). OUP.

Cummins, R. (2010). "How Does It Work ?" versus "What Are the Laws ?": Two Conceptions of Psychological Explanation. In *The World in the Head*. Oxford University Press. https://doi.org/10.1093/acprof:osobl/9780199548033.003.0016

Dames, H., Musfeld, P., Popov, V., Oberauer, K., & Frischkorn, G. T. (2024). Responsible Research Assessment Should Prioritize Theory Development and Testing Over Ticking Open Science Boxes. *Meta-Psychology*, *8*. https://doi.org/10.15626/MP.2023.3735

Davis-Stober, C. P. (2021, August 17). *Alien in a can.* https://www.davis-stober.com/post/alien-in-a-can

Davis-Stober, C. P., Sarafoglou, A., Aczel, B., Chandramouli, S. H., Errington, T. M., Field, S. M., ... & Gundersen, O. E. (2025). How can we make sound replication decisions?. *Proceedings of the National Academy of Sciences*, *122*(5), e2401236121. https://doi.org/10.1073/pnas.2401236121

Deutsch, D. (2011). *The beginning of infinity: Explanations that transform the world*. penguin uK.

Eronen, M. I., & Bringmann, L. F. (2021). The theory crisis in psychology: How to move forward. *Perspectives on Psychological Science, 16*(4), 779–788. https://doi.org/10.1177/1745691620970586

Fiedler, K. (2017). What constitutes strong psychological science? The (neglected) role of diagnosticity and a priori theorizing. *Perspectives on Psychological Science*, 12(1), 46-61. https://doi.org/10.1177/1745691616654458

Field, S. M., Volz, L., Kaznatcheev, A., & van Dongen, N. (2024). Can a good theory be built using bad ingredients?. Computational Brain & Behavior, 7(4), 608-615. https://doi.org/10.1007/s42113-024-00220-w

Forstmann, B. U., Wagenmakers, E.-J., Eichele, T., Brown, S., & Serences, J. T. (2011). Reciprocal relations between cognitive neuroscience and formal models. *Trends in Cognitive Sciences, 15*(6), 272–279. https://doi.org/10.1016/j.tics.2011.04.002

Forscher, B. K. (1963). Chaos in the brickyard. *Science, 142*(3590), 339.

Frankenhuis, W. E., Borsboom, D., Nettle, D., & Roisman, G. I. (2023). Formalizing theories of child development: Introduction to the special section. Child Development, 94, 1425–1431. https://doi.org/10.1111/cdev.14020

Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic ambiguity in the social sciences. *Social Psychological Bulletin*, *18*, 1-25. https://doi.org/10.32872/spb.9923

Fried, E. I., et al. (2023). Lack of theory building and testing impedes progress in the factor and network literature. *Psychological Inquiry, 31(4)*, 271-288. https://doi.org/10.1080/1047840X.1853461

Gärtner, A., Leising, D., Freyer, N., Musfeld, P., Lange, J., & Schönbrodt, F. D. (2025). *Responsible Research Assessment II: A specific proposal for hiring and promotion in psychology*. PsyArXiv. https://doi.org/10.31234/osf.io/5yexm_v2

Gershman, S. J. (2019). How to never be wrong. Psychonomic bulletin & review, 26(1), 13-28. https://doi.org/10.3758/s13423-018-1488-8

Guest, O., & Martin, A. E. (2021). How computational modeling can force theory building in psychological science. *Perspectives on Psychological Science, 16*(4), 789–810. https://doi.org/10.1177/1745691620970585

Glöckner, A., & Betsch, T. (2011). The empirical content of theories in judgment and decision making: Shortcomings and remedies. Judgment and Decision Making, 6(8), 711-721. https://doi.org/10.1017/S1930297500004149

Glöckner, A., Fiedler, S., & Renkewitz, F. (2018). Belastbare und effiziente Wissenschaft: Strategische Ausrichtung von Forschungsprozessen als Weg aus der Replikationskrise [Sound and efficient science: a strategic alignment of research processes as way out of the replication crisis]. Psychologische Rundschau, 69(1), 22-36. https://doi.org/10.1026/0033-3042/a000384

Glöckner, A., Gollwitzer, M., Hahn, L., Lange, J., Sassenberg, K., & Unkelbach, C. (2024). Quality, Replicability, and Transparency in Research in Social Psychology. Social Psychology. https://doi.org/10.1027/1864-9335/a000548

Glöckner, A., & Fiedler, S. (2025). How to Improve Theory Specification in Psychology: A Review and Five Policy Suggestions. Zeitschrift Für Psychologie, 233(4), 224–239. https://doi.org/10.1027/2151-2604/a000599

Haig, B. D. (2018). An abductive theory of scientific method. In *Method matters in psychology: Essays in applied philosophy of science* (pp. 35-64). Cham: Springer International Publishing.

Hale, J., Hastings, J., West, R. et al. (2020). An ontology-based modelling system (OBMS) for representing behaviour change theories applied to 76 theories. *Wellcome Open Research*, 5:177. https://doi.org/10.12688/wellcomeopenres.16121.1

Harris, R. J. (1976). The uncertain connection between verbal theories and research hypotheses in social psychology. *Journal of Experimental Social Psychology*, *12*(2), 210–219. https://doi.org/10.1016/0022-1031(76)90071-8

Haslbeck, J., Ryan, O., Robinaugh, D. J., Waldorp, L. J., & Borsboom, D. (2022). Modeling psychopathology: From data models to formal theories. *Psychological Methods*, *27*(6), 930. https://doi.org/10.1037/met0000303

Hosseinichimeh, N., Rahmandad, H., & Wittenborn, A. K. (2015). Modeling the hypothalamus-pituitary-adrenal axis: A review and extension. Mathematical Biosciences, 268, 52-65. https://doi.org/10.1016/j.mbs.2015.08.004

Hosseinichimeh, N., Wittenborn, A. K., *Rick, J., Jalali, M. S., & Rahmandad, H. (2018). Modeling and estimating the feedback mechanisms among depression, rumination, and stressors in adolescents. PLOS ONE, 13(9), e0204389. https://doi.org/10.1371/journal.pone.0204389

Jonas, E., & Kording, K. P. (2017). Could a neuroscientist understand a microprocessor?. PLoS computational biology, 13(1), e1005268. https://doi.org/10.1371/journal.pcbi.1005268

Kellen, D. (2019). A model hierarchy for psychological science. *Computational Brain & Behavior*, *2*(3), 160-165. https://doi.org/10.1007/s42113-019-00037-y

Kovacs, Kristof, and Andrew RA Conway. "What is IQ? Life beyond “general intelligence”." *Current Directions in Psychological Science* 28, no. 2 (2019): 189-194. https://doi.org/10.1177/0963721419827275

Kruglanski, A. W. (2001). That "vision thing": The state of theory in social and personality psychology at the edge of the new millennium. Journal of Personality and Social Psychology, 80(6), 871-875. https://doi.org/10.1037/0022-3514.80.6.871

Lange, J., Freyer, N., Musfeld, P., Schönbrodt, F., & Leising, D. (2025). A Checklist for Incentivizing and Facilitating Good Theory Building. *Zeitschrift Für Psychologie*, 2151-2604/a000604. https://doi.org/10.1027/2151-2604/a000604

Leising, D., Thielmann, I., Glöckner, A., Gärtner, A., & Schönbrodt, F. (2022). Ten Steps Toward a Better Personality Science – How Quality May Be Rewarded More in Research Evaluation. Personality Science, 3(1), 1-44. https://doi.org/10.5964/ps.6029

Lykken, D.T. (1991) What's Wrong with Psychology Anyway?. In D.Cicchetti and W.M. Grove. (Eds.). *Thinking Clearly about Psychology. Volume 1: Matters of Public Interest.* University of Minnesota Press.

Maier, M., van Dongen, N., & Borsboom, D. (2024). Comparing theories with the Ising model of explanatory coherence. *Psychological Methods*, 29(3), 519–536. https://doi.org/10.1037/met0000543

McPhetres, J., Albayrak-Aydemir, N., Barbosa Mendes, A., Chow, E. C., Gonzalez-Marquez, P., Loukras, E., Maus, A., O'Mahony, A., Pomareda, C., Primbs, M. A., Sackman, S. L., Smithson, C. J. R., & Volodko, K. (2021). A decade of theory as reflected in Psychological Science (2009–2019). *PLOS ONE*, *16*(3), e0247986. https://doi.org/10.1371/journal.pone.0247986

Meadows, D. (2008). *Thinking in systems: International bestseller*. Chelsea Green Publishing.

Meehl, P. E. (1967). Theory-testing in psychology and physics: A methodological paradox. *Philosophy of Science, 34*(2), 103–115. https://doi.org/10.1086/288135

Meehl, P. E. (1978). Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology. *Journal of Consulting and Clinical Psychology, 46*(4), 806–834. https://doi.org/10.1037/0022-006X.46.4.806

Meehl, P. E. (1990a). Why summaries of research on psychological theories are often uninterpretable. *Psychological Reports, 66*(1), 195–244. https://doi.org/10.2466/pr0.1990.66.1.195

Meehl, P. E. (1990b). Appraising and Amending Theories: The Strategy of Lakatosian Defense and Two Principles that Warrant It. Psychological Inquiry, 1(2), 108–141. https://doi.org/10.1207/s15327965pli0102_1

Muthukrishna, M., & Henrich, J. (2019). A problem in theory. *Nature Human Behaviour, 3*(3), 221–229. https://doi.org/10.1038/s41562-018-0522-1

Nak, J., Volz, L.,Verbrugh, V., Borsboom, D., & van Dongen, N.N.N. (2025). The PsychoModels database—Facilitating the exchange of formal models. *Zeitschrift für Psychologie* https://doi.org/10.1027/2151-2604/a000601

Oberauer, K., & Lewandowsky, S. (2019). Addressing the theory crisis in psychology. *Psychonomic Bulletin & Review, 26*(5), 1596–1618. https://doi.org/10.3758/s13423-019-01645-2

Oude Maatman, F. J. W. (2025). Psychology's Theory Crisis, and Why Formal Modelling Cannot Solve It. *Meta-Psychology*. https://doi.org/10.31234/osf.io/puqvs

Popov, V. (2023, January 3). If God Handed Us the Ground-Truth Theory of Memory, How Would We Recognize It?. https://doi.org/10.31234/osf.io/ay5cm

Popper, K. (2019). *Conjectures and refutations: The growth of scientific knowledge*. routledge.

Reisenzein, R. (2021). Tasks for a theoretical psychology of emotion. Cognition and Emotion, 36(2), 171-187. https://doi.org/10.1080/02699931.2021.1992356

Robinaugh, D. J., Haslbeck, J. M. B., Waldorp, L., Kossakowski, J., Fried, E. I., Millner, A., McNally, R. J., van Nes, E. H., Scheffer, M., Kendler, K. S., & Borsboom, D. (2024). Advancing the network theory of mental disorders: A computational model of panic disorder. *Psychological Review*, 131(6), 1482-1508. https://doi.org/10.1037/rev0000515

Robinaugh, D. J., Haslbeck, J. M., Ryan, O., Fried, E. I., & Waldorp, L. J. (2021). Invisible hands and fine calipers: A call to use formal theory as a toolkit for theory construction. *Perspectives on Psychological Science*, *16*(4), 725-743. https://doi.org/10.1177/1745691620974697

Rozin, P. (2009). What kind of empirical research should we publish, fund, and reward? A different perspective. *Perspectives on Psychological Science, 4*(4), 435–439. https://doi.org/10.1111/j-6924.2009.01151.x

Scheel, A. M., Tiokhin, L., Isager, P. M., & Lakens, D. (2021). Why hypothesis testers should spend less time testing hypotheses. *Perspectives on Psychological Science*, *16*(4), 744-755. https://doi.org/10.1177/1745691620966795

Schönbrodt, F. D., Gärtner, A., Frank, M., Gollwitzer, M., Ihle, M., Mischkowski, D., Phan, L. V., Schmitt, M., Scheel, A. M., Schubert, A.-L., Steinberg, U., & Leising, D. (2025). *Responsible Research Assessment I: Implementing DORA and CoARA for hiring and promotion in psychology*. PsyArXiv. https://doi.org/10.31234/osf.io/rgh5b_v2

Smaldino, P. E. (2020). How to translate a verbal theory into a formal model. *Social Psychology*. https://doi.org/10.1027/1864-9335/a000425

Smaldino, P. E., & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science*, *3*(9), 160384. https://doi.org/10.1098/rsos.160384

Tiokhin, L. (2021, August 15). *Theories and models are not the only fruit. Medium.* https://leotiokhin.medium.com/theories-and-models-are-not-the-only-fruit-a05c7cf188f6

van Dongen, N., van Bork, R., Finnemann, A., Haslbeck, J. M. B., van der Maas, H. L. J., Robinaugh, D. J., de Ron, J., Sprenger, J., & Borsboom, D. (2025a). Productive explanation: A framework for evaluating explanations in psychological science. *Psychological Review*, 132(2), 311–329. https://doi.org/10.1037/rev0000479

van Dongen, N., Finnemann, A., de Ron, J., Tiokhin, L., Wang, S., Algermissen, J., ... & Borsboom, D. (2025b). Practicing theory building in a many modelers hackathon: A proof of concept. Meta-Psychology, 9. https://doi.org/10.15626/MP.2023.3688

Van Lissa, C. J., Peikert, A., Ernst, M., Dongen, N. V., Schönbrodt, F., & Brandmaier, A. M. (2025). To be FAIR: Theory Specification Needs an Update. OSF. https://doi.org/10.31234/osf.io/t53np_v3

van Rooij, I. (2019, January 18). *Psychological science needs theory development before preregistration. Psychonomic Society Featured Content.* https://featuredcontent.psychonomic.org/psychological-science-needs-theory-development-before-preregistration

van Rooij, I. (2022). Psychological models and their distractors. Nature Reviews Psychology, 1(3), 127-128. https://doi.org/10.1038/s44159-022-00031-5

van Rooij, I., & Blokpoel, M. (2020). Formalizing verbal theories. Social Psychology. https://doi.org/10.1027/1864-9335/a000428

Vaidyanathan, U., Vrieze, S. I., & Iacono, W. G. (2015). The power of theory, research design, and transdisciplinary integration in moving psychopathology forward. *Psychological Inquiry*, *26*(3), 209-230. https://doi.org/10.1080/1047840X.2015.1015367

Wittenborn, A. K., Rahmandad, H., Rick, J., & Hosseinichimeh, N. (2016). Depression as a systemic syndrome: Mapping the feedback loops of major depressive disorder. Psychological Medicine, 46, 551-562. https://doi.org/10.1017/S0033291715002044

Wittenborn, A. K., & Hosseinichimeh, N. (2022). Exploring personalized psychotherapy for depression: A systems dynamics approach. PLOS ONE, 17(10), e0276441. https://doi.org/10.1371/journal.pone.0276441